

**Review of Manufacturing Workplace Burnout Risk Factors Across Parts of the Globe,  
Including Physical and Financial Costs**

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## Introduction

Workplace burnout is a global phenomenon recognized heavily in healthcare, education, and social work. However, widespread surveys have reflected considerable burnout in the manufacturing industry (TNO & CBS, 2018; Eurofound, 2022). This literature review specifically observed burnout risk factors in the manufacturing industry across parts of the globe. Physical and financial costs were also summarized.

Google Scholar and EBSCO Databases were searched for articles in the English language between January 2012 and February 2023. Search terms mandated requirements within the title only: to include spelling variations of “burnout” and “manufacturing” while excluding out-of-scope industries and occupations. The combined search offered 30 unique results, of which eight were published during COVID-19 (post-January 2020.) Articles were selected if they could provide enough information regarding the tools used and analyses conducted.

The top two burnout measuring scales and how studies have used them to explore relationships with other themes were reviewed. Some studies were weakened by statistical analysis omissions, specific approaches to language translations, or burnout summarization errors. Powerful themes emerged, which prompted an additional internet search for comparative workplace observations at country, continental, and global scales (Eurofound, 2022; Gallup, 2022; TNO & CBS, 2018). Rough magnitudes estimating burnout cost were only available at these higher levels from elder sources.

## Burnout Factor Definition and Recognition

Developing a formal definition for such a complex state has been challenging. Psychologist Herbert Freudenberger was credited for coining “burnout” due to his own

experience with workplace exhaustion by 1974. Consequentially, the first scientifically developed measure of burnout targeted human service professionals in 1981 (Maslach & Jackson, 1981). By 2019, the World Health Organization aligned with the Maslach Burnout Inventory's definition. The MBI reframed burnout as a syndrome consisting of *all* three parts: emotional exhaustion, cynicism, and professional efficacy. The combined profile - *rather than the summation* - of high emotional exhaustion and cynicism with low professional efficacy scores signaled a *likelihood* of burnout. It was not a medical diagnostic tool.

The MBI-GS is the General Survey version of the questionnaire and was the most popular resource for observing manufacturing professionals' burnout in this review. The MBI series is the world's leading burnout measurement (Maslach & Leiter, 2021).

Researchers have other options if not for the MBI-GS scale ("Valid and Reliable," 2022). Among them is the Burnout Measure, developed by Pines and Aronson in 1988. Malach-Pines (2005) developed a **shortened version** (BM-SV) in response to application feedback for a smaller, quicker, and easier-to-understand instrument. In contrast to the MBI-GS, the BM-SV consists of fewer statements (10 versus 16) and focuses only on one burnout subcomponent: emotional exhaustion. Both use seven-point Likert scales to remark the frequency of experiencing each statement. Each requires five to ten minutes to complete.

## Critical Considerations for Reliable Data and Outcomes

The most accurate articles considered the following components in this section. Each subsection includes a review of any noteworthy article performances.

## Properly Merged Measurement Tools

Burnout scales were combined with several other tools to investigate relationships with various subjects, as previewed in Table 1. Subjects were diverse and clarified personal feelings, co-worker relationships, job quality, the working environment, and others. Few studies covered the same subject. When they did, different measurements were used.

When combining questions, trustworthy articles included checks for reliability and validity. For a deeper exploration of methods, see Appendix A. See Appendix B for the full version of Table 1.

*Table 1*

*Selection of Additional Subjects and How Well They Were Merged*

| Subject                                           | Paired Tool                                                                                          | Paired Tool |                    | Burnout Measure | Article Source              |
|---------------------------------------------------|------------------------------------------------------------------------------------------------------|-------------|--------------------|-----------------|-----------------------------|
|                                                   |                                                                                                      | Reliability | Construct Validity |                 |                             |
| Turnover Intention<br>(a worker has not left yet) | Landau & Hammer – 3 items                                                                            | Yes         | Yes                | BM-SV           | Santhanam & Srinivas, 2019  |
|                                                   | Brough & Frame - 3 items                                                                             | Yes         | Low threshold      | MBI-GS          | Marshall & Stephenson, 2020 |
| Social Support                                    | Karasek & Theorell – four items, a five-point scale                                                  | Yes         | Yes                | MBI-GS          | Chen et al., 2018           |
|                                                   | Van Veldhoven & Meijman, – six items, five-point Likert scale, a subset of psychosocial work demands | Yes         | Missing            | MBI-GS          | Hulsegge et al., 2020       |
| Quality of Working Environment                    | OECD– condensed by article source authors                                                            | Missing     | Missing            | Unknown         | Ipsos NV, Belgium, 2022     |
| Success in Business                               | Environmental, Social, and Governance criteria – unknown details                                     | Unknown     | Unknown            | Unknown         | Gallup, 2022                |

Marshall and Stephenson (2020) applied their turnover intention study to an electronics company in South Africa with several challenges. Their factorial analysis led to a consolidation of exhaustion and cynicism, which could indicate that the questionnaire was inappropriate for

that audience (Maslach et al., 1997). They also accepted a low threshold  $|r| \geq 0.3$ , when  $|r| \geq 0.5$  is generally a more accepted threshold for construct validity. Continental and global surveys need more information for their statistical methods.

### Care with Language Translations

For surveys to be effective, they must be understood. The MBI-GS was written and validated in English. Any translations require permission and re-validation because a translation may not perfectly reflect the same meaning or sentiment (Maslach et al., 1997). If the study translated a survey for the first time, the results were potentially not as trustworthy. These considerations are summarized in Table 2 below.

Marshall and Stephenson (2020) used a pilot run to vet understanding before formal execution. The 27+ country survey in Europe also greatly benefited from catching significant errors – language and non-language related - from their pilot run (Ipsos NV, Belgium, 2022).

*Table 2*

*Summary of Articles Whose Survey Language Was Translated Via the Article Author's Efforts*

| Article Source                           | Language                          | Approach                                                                                                                                                                                                            |
|------------------------------------------|-----------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Santhanam & Srinivas, 2019               | Unknown - India                   | Translated by at least three professional translators                                                                                                                                                               |
| Marshall & Stephenson, 2020              | Afrikaans                         | Coordination between an English-speaking statistician, a company-based English-speaking manager, and two Afrikaans-speaking worker union representatives of undeclared proficiencies or native languages            |
| Eurofound, 2022; Ipsos NV, Belgium, 2022 | Fifty-five European languages     | It was translated by two independent professional translators, native to each language. Results were reviewed by each country's interview agency project manager with native language researcher support as needed. |
| Gallup, 2022                             | Global languages of 160 countries | Not declared                                                                                                                                                                                                        |

## **Accurately Relaying Burnout Results**

While the MBI-GS is the gold standard for measuring burnout, it leaves room for desire (Maslach et al., 1997). The three-part scores are not as clean as a single summarized value. From an applications perspective, researchers, companies, and governments strive to determine summarized burnout rates and their severities. Consequentially, oversimplified results needed to be more accurate. Added scores (Hulsegge, 2020) or arbitrarily splitting results into high, medium, and low burnout severity (Marshall & Stephenson, 2020) were observed.

The MBI collaborators reminded researchers to combine the MBI-GS with other measurement tools and include areas for open respondent commentary to reveal deeper insights (Maslach & Leiter, 2021.) Additionally, they emphasized that having different combinations of results across the three categories represents five different profiles: burnout, overextended, ineffective, disengaged, and engaged. Each profile requires better understanding, but the approach to support each would presumably differ.

## **Result Comparisons**

The four regional articles with relatively more reliable data are discussed in detail and compared to global findings.

## **Regional Highlights**

As seen in Table 4, the manufacturing industry was predominantly male and young-to-middle-aged. Blue-collar positions were likely to have high-school or lower levels of education.

Table 3

*Demographics Summary From the Strongest Regional Studies*

| Source                      | Location         | Occupation level                     | Gender Predominance  | Age (mean +/- standard deviation) in years | Education                                                    | Weekly working hours |
|-----------------------------|------------------|--------------------------------------|----------------------|--------------------------------------------|--------------------------------------------------------------|----------------------|
| Chen et al., 2018           | China – Southern | Mostly blue-collar with white-collar | Did not disclose     | ~26 +/- 4.7                                | Low – 10.5 +/- 1.9 years                                     | Could not disclose   |
| Santhanam & Srinivas, 2019  | India            | Blue-collar                          | Slightly male (~55%) | 31 +/- 12.03                               | Low – 97% without a high school diploma                      | Missing              |
| Valadez-Torrez et al., 2017 | Mexico           | White-collar                         | Strongly male (~69%) | 37 +/- ?                                   | Moderate - ~62% Bachelor's                                   | >42 (~80%)           |
| Wang et al., 2017           | China – Northern | Missing                              | Strongly male (~76%) | 35.44 +/- 9.47                             | Moderate - ~40% junior college and ~40% Bachelor's or higher | <40 (~69%)           |

All studies found a connection between emotional exhaustion and adverse workplace outcomes. India's analysis could not extend its relationship to burnout because it did not use a full burnout scale (Santhanam & Srinivas, 2019). Despite this, studies in Northern China, Southern China, and Mexico supported the three-dimensional definition of burnout.

Factors associated with higher burnout risk were identified. In Northern China, lower education, earnings, and tenure at a company are concerns (Wang et al., 2017). Likewise, longer weekly working hours are a factor, although the company was very concerned about sharing the

figures publicly. In Southern China, lower autonomy was the critical factor in their focused study (Chen et al., 2017).

Factors that mediated adverse outcomes were also discovered. Having good social support – particularly from a supervisor - was the most important lesson from these studies. Additionally, happiness was associated with engaged employees and buffered the association with turnover intention (Santhanam & Srinivas, 2019). Possession of psychological resources (such as confidence, hope, good self-esteem, and resilience) majorly supported well-being.

A few other interesting observations were made:

- 1) Elder workers exhibited less emotional exhaustion (Wang et al., 2017).
- 2) Employees who expended much effort tended to have higher levels of cynicism (Wang et al., 2017).
- 3) Formal reflexivity training (where coworkers openly analyze their behavior, relate it to successes and failures, and are actively involved in decision-making processes) with low-tenured teams saw a drop in performance (Chen et al., 2017).

## **Regional-Global Highlights**

Although mass-scale surveys lacked some methodological information, they could fairly compare results from vast regions. They also had the advantage of observing changes over time.

Gallup's and Eurofound's working condition surveys during the onset of COVID-19 took similar care measures to ensure reliability (Gallup, 2022; Eurofound, 2022). Random selection, data weighting against unequal selection probability, and checks against national demographics were standard to reduce bias. Their eligibility requirements were also similar: a minimum age of 15 or 16, respectively. Language translations were a heavy component of both sources.

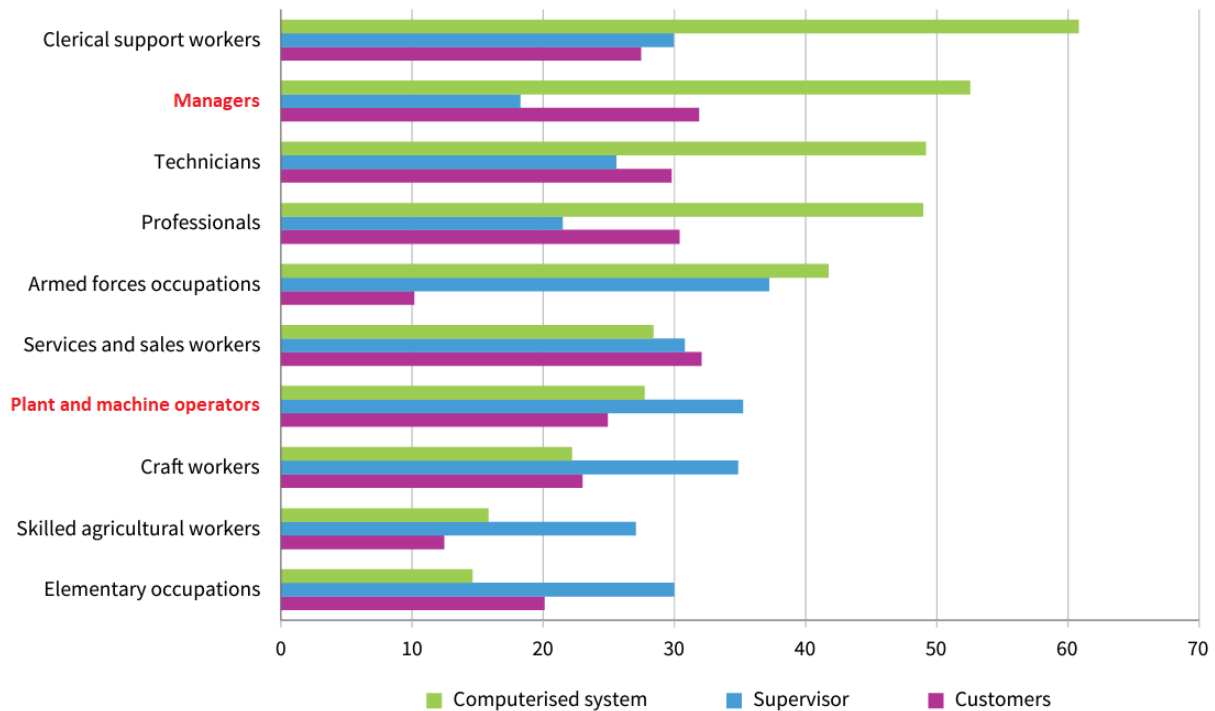


Key differences in these sources stemmed from Gallup's world outreach of 160 countries versus Eurofound's EU-and-proximity outreach of 36 countries. Gallup also had a more substantial consideration of life and family factors than Eurofound and did not group data by occupation. Gallup excluded self-employed workers and always used a mixture of face-to-face and telephone interviews. In contrast, Eurofound was purely telephone-based.

Both broad reviews recognized burnout and referred to it using MBI's terms - Eurofound even called the MBI out by name. Both sources emphasized that engagement to reduce burnout was a consistent theme from their past findings. The intense influence that the supervisor has on an employee was called out and supported the earlier regional observations. Eurofound elaborated that this relationship was exasperated in the manufacturing industry; Supervisors primarily influence job duties and, therefore, an individual's autonomy. Figure 1 illustrates this observation in the context of other occupations. Further parallels to regional findings occur with the link between engagement and job quality; those with low engagement were more likely to be highly strained. On par with Valadez-Torrez et al., Eurofound reported that plant and machine workers had the second-highest occupational health and safety risk in the EU27 – 11% higher than the entire sample's average.

**Figure 1**

*Employees Influenced to a Large Extent by a Computerized System, a Supervisor, and Customers, by Occupation, EU27 (%) (Eurofound, 2022, p. 60)*



*Note. 1 The literature review author modified manufacturing-related occupations to bold red font. From Eurofound, 2022. Copyright 2022 by Eurofound.*

Demographically, Eurofound's data resolution combined mining, construction, and transport with manufacturing but reported a comparable 71% male predominance versus the 66.6% average of regional observations. Additionally, *managers* across all occupations in Eurofound's data indicated an even higher 87% male predominance. Though not a direct comparison, occupation level showed several contextual differences that merit comparing blue- and white-collar positions.

Factors yet to be mentioned but important in the global reviews included work-life balance, presence and age of children, marital status, and weekly working hours. Gallup brought in unique considerations like daily emotions (anger and sadness), the country's environmental

impact, and perceptions of corruption. Such factors could be incorporated into future regional studies.

## **Cost of Burnout**

The cost of burnout has been difficult to calculate, yet it is a crucial factor dictating whether a manufacturing company pursues its reduction. Burnout's broader definition overlaps with chronic stress, anxiety, depression, and other states (Hassard et al., 2014). Without the ability to distinguish burnout using a formal diagnosis (World Health Organization, 2019), the combined cost is one workable perspective.

The magnitudes were high: the Netherlands alone spent an estimated 1.8 billion euros in 2017 to cover absenteeism from stress (TNO & CBS, 2018). An older 2005 report collected by Hassard et al. (2014) reported 1.3 billion euros in absenteeism for the Netherlands. The source expanded upon additional 1.7 billion euros in disability benefits and 1.02 billion euros in medical costs that same year.

“Presenteeism” is when employees work when ill. This is an expense for employers as well. In 2011, Germany paid 2,399 euros per employee to cover presenteeism. Eurofound's 2022 results indicated that the established working culture hosts 44% of on-location production workers with three or more health problems.

Regional and global surveys touched on (missed) productivity gains when workers suffer burnout. A 25% boost in productivity was observed in tenured teams who received formal reflexivity training (Chen et al., 2017). Gallup (2022) observed 23% higher profit from engaged workers versus actively disengaged ones.

Physically, burnout has been observed to impart both short- and long-term adverse effects directly on the worker (van Dijk et al., 2020). Short-term effects included poor sleep quality. Mid- and long-term effects had many cognitive impairments, particularly affecting the central executive functions (Blix et al., 2013). Such functions enable a person to focus, maintain task goals, make decisions, and recall memories. Between the caudate, putamen, and mesial frontal cortex, processes such as learning and motor control, reward, cognitive function, addiction, and decision-making are executed (Blix et al., 2013; Savic, 2013). Prolonged stress has shown excess thinning and gray matter shrinkage in these brain parts.

The changes could explain why burnout leads to safety accidents. A predictive algorithm by a New Zealand health and safety software company associated poor safety performance with psychosocial factors. Those with boosted autonomy and social support had lower burnout and higher safety performance (van Marrewijk, 2022).

## Conclusion

The Maslach Burnout Inventory – General Survey is the gold standard for measuring workplace burnout in the manufacturing industry. Though other scales exist, this one considers burnout’s complete profile: the combination of emotional exhaustion, cynicism, and low professional efficacy. The scale must be combined with other scales to ascertain relationships between potential causes or outcomes. Reliability, construct validity, and any necessary language translations should be reviewed carefully before administering the compiled survey. These considerations may require adding, removing, or adjusting statements to regain survey trustworthiness. Summarizations should not condense burnout into a single metric.

Four region-specific articles navigated these considerations well. They reflect a male-dominated industry, where blue-collar workers are far less educated than white-collar workers. The educational deficit, lower earnings, and lower tenure at a company correlated with higher burnout risk for blue-collar workers. Combined with continental and global information, the theory of a very close supervisor-worker relationship may explain why social support is critical for alleviating burnout. Happiness and strong psychological resources such as confidence and self-esteem could protect a worker's mind and body.

If companies do not change their workplace culture, burnout will continue to spread. The status quo is costing countries billions of euros annually, and preventable safety incidents have maimed or lost an indeterminate number of lives. As a call to action, researchers should investigate the success of different burnout alleviation strategies in the manufacturing industry. The findings in this review allude that a few different supervisor-focused social support strategies monitored over time may be particularly successful. Special considerations include age, tenure with the company, tenure with their closest team, and weekly working hours. Cultural perceptions of mental health (whether stigmatized or not) may be hurdles to navigate.

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## **Appendix A**

### **Explanation of Reliability and Construct Validity**

The resulting list should be checked for reliability and validity whenever questions are compiled – whether creating, modifying, or combining pre-existing validated questionnaires. Structural equation modeling is one means to check for reliability and differential validity.

#### **Reliability**

Each question should associate with a single variable or concept, and groups of questions measuring the same concept should correlate. Several methods exist to do this, such as:

- 1) Test-retest reliability coefficients per subscale (concept) show stability over time with strong statistical significance ( $<0.001$ ) (Maslach et al., 1997).
- 2) Internal consistency by Cronbach's alpha ( $\alpha$ ) coefficient is acceptable with values above 0.70. Only the global analyses were unclear on whether this was used (Gallup, 2022; Ipsos NV, Belgium, 2022).

#### **Construct Validity**

Two perspectives are necessary to confirm construct validity: convergent and discriminant validity.

Convergent validity occurs when two methods or questions measure a concept, resulting in similar scores. Confirmation can be achieved by any of the following, ideally through multiple ways:

- 1) comparing an individual's self-assessed results to those scored by a group of co-workers who knew that person very well (Maslach & Jackson, 1981)

- 2) correlating questionnaire scores to specific job characteristics already associated with burnout (such as the number of clients or proportion of time in direct contact with clients) (Maslach & Jackson, 1981)
- 3) correlating questionnaire scores to previously validated personal burnout outcomes (such as feeling dissatisfied with available growth opportunities or doubting that their work is meaningful) (Maslach & Jackson, 1981)
- 4) estimating the average variance extracted (AVE) (Valadez-Torrez et al., 2017)

On the other hand, discriminant validity is proof that two tests (or questions) that should be distinct actually are. It is beneficial in surveys because authentic personal responses may be at odds with socially or professionally acceptable ideals (Maslach & Jackson, 1981).

- 1) Comparing a new questionnaire to the Crowne-Marlowe (1964) Social Desirability Scale or similar
- 2) Computing cross-loadings, with p-value  $\leq 0.05$  and loads  $\geq 0.50$  (Valadez-Torrez et al., 2017)

Both convergent and discriminant validities should be proven before conducting the survey. Calculating a correlation coefficient like Pearson's  $r$  can assess both types of validity at once. Values of  $|r| > 0.50$  are generally considered acceptably correlated, while unrelated concepts should be relatively lower and closer to 0. Most articles did this.

## Appendix B

### Detailed Summary Tables of Subject Reliability and Construct Validity

Table 4

*Full Summary of Additional Subjects and How Well They Were Merged in Their Respective Studies*

| Subject                                                                                          | Paired Tool                                                                                          | Paired Tool |                    | Burnout Measure | Article Source              |
|--------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------|-------------|--------------------|-----------------|-----------------------------|
|                                                                                                  |                                                                                                      | Reliability | Construct Validity |                 |                             |
| Turnover Intention<br>(a worker has not left yet)                                                | Landau & Hammer – 3 items                                                                            | Yes         | Yes                | BM-SV           | Santhanam & Srinivas, 2019  |
|                                                                                                  | Brough and Frame–<br>Modified by article source authors - 3 items                                    | Yes         | Low threshold      | MBI-GS          | Marshall & Stephenson, 2020 |
| Musculoskeletal Complaints<br>(location, frequency, intensity, and likelihood to seek treatment) | Body Map Assessment                                                                                  | Yes         | Yes                | MBI-GS          | Valadez-Torrez et al., 2017 |
| Job Content<br>(psychosocial factors at work)                                                    | Job Content Questionnaire–<br>27 items, four-point Likert scale                                      | Yes         | Yes                | MBI-GS          | Valadez-Torrez et al., 2017 |
| Reflexivity                                                                                      | Carter & West– five items, seven-point Likert scale                                                  | Yes         | Yes                | MBI-GS          | Chen et al., 2018           |
| Social Support                                                                                   | Karasek & Theorell – four items, five-point scale                                                    | Yes         | Yes                | MBI-GS          | Chen et al., 2018           |
|                                                                                                  | Van Veldhoven & Meijman, – six items, five-point Likert scale, a subset of psychosocial work demands | Yes         | Missing            | MBI-GS          | Hulsegge et al., 2020       |
| Control                                                                                          | Job Control – 22 items, five-point Likert scale                                                      | Yes         | Yes                | MBI-GS          | Chen et al., 2018           |
|                                                                                                  |                                                                                                      |             |                    |                 |                             |
|                                                                                                  |                                                                                                      |             |                    |                 |                             |
|                                                                                                  |                                                                                                      |             |                    |                 |                             |
|                                                                                                  |                                                                                                      |             |                    |                 |                             |
|                                                                                                  |                                                                                                      |             |                    |                 |                             |

| Subject                              | Paired Tool                                                                                                | Paired Tool |                    | Burnout Measure | Article Source             |
|--------------------------------------|------------------------------------------------------------------------------------------------------------|-------------|--------------------|-----------------|----------------------------|
|                                      |                                                                                                            | Reliability | Construct Validity |                 |                            |
| Role Overload                        | Qualitative Role Overload – five items, seven-point Likert scale                                           | Yes         | Yes                | MBI-GS          | Chen et al., 2018          |
| Engagement                           | Saks' Employee Engagement                                                                                  | Yes         | Yes                | BM-SV           | Santhanam & Srinivas, 2019 |
|                                      | Custom-made by article source author "Q12" – 12 items, five-point Likert scale                             | Unknown     | Unknown            | Unknown         | Gallup, 2022               |
| Life Evaluation                      | Self-Anchoring Striving Scale – modified by article source author. Two items, scale ranges zero to 10      | Unknown     | Unknown            | Unknown         | Gallup, 2022               |
| Success in Business                  | Environmental, Social, and Governance criteria – unknown details                                           | Unknown     | Unknown            | Unknown         | Gallup, 2022               |
| Job Market                           | Custom-made by article source author – 3 items, fixed responses                                            | Unknown     | Unknown            | Unknown         | Gallup, 2022               |
| Happiness                            | Oxford Happiness Questionnaire                                                                             | Yes         | Yes                | BM-SV           | Santhanam & Srinivas, 2019 |
| Distress                             | Four-Dimensional Symptom Questionnaire – 16 items, 5-point Likert scale, distress subscale only.           | Missing     | Missing            | MBI-GS          | Hulsegge et al., 2020      |
| Work Schedule Satisfaction           | Custom-made by article source author – 1 item, 5-point Likert scale                                        | N/A         | N/A                | MBI-GS          | Hulsegge et al., 2020      |
| Impact of Shift Work on Private Life | Custom-made by article source author – 1 item, 5-point Likert scale                                        | Yes         | Missing            | MBI-GS          | Hulsegge et al., 2020      |
| Work Pressure                        | Van Veldhoven & Meijman, 1994 – five items, five-point Likert scale, a subset of psychosocial work demands | Yes         | Missing            | MBI-GS          | Hulsegge et al., 2020      |
| Quality of Working Environment       | OECD – condensed by article source authors                                                                 | Missing     | Missing            | Unknown         | Ipsos NV, Belgium, 2022    |
|                                      |                                                                                                            |             |                    |                 |                            |
|                                      |                                                                                                            |             |                    |                 |                            |

| Subject                  | Paired Tool                                                                                                      | Paired Tool |                    | Burnout Measure | Article Source    |
|--------------------------|------------------------------------------------------------------------------------------------------------------|-------------|--------------------|-----------------|-------------------|
|                          |                                                                                                                  | Reliability | Construct Validity |                 |                   |
| Occupational Stress      | Effort-Reward Imbalance model – 23 items, includes all subsets (extrinsic effort, reward, and over-commitment)   | Yes         | Yes                | MBI-GS          | Wang et al., 2017 |
| Psychological Capacity   | Psychological Capital Questionnaire – 24 items, includes all subsets (self-efficacy, hope, resilience, optimism) | Yes         | Yes                | MBI-GS          | Wang et al., 2017 |
| Self Esteem              | Rosenberg Self-Esteem Scale – 10 items, four-point Likert scale                                                  | Yes         | Yes                | MBI-GS          | Wang et al., 2017 |
| Psychological Well-being | Flourishing Scale (Sumi, 2014) – 8 items, numerical scale ranging from eight to 56                               | Yes         | Yes                | MBI-GS          | Wang et al., 2017 |